Run report: generator-effect E4, 2026-10-19T22:15Z

Descriptive record of the E4 ecbench run. The interpretation is in `../../E0_E4.md`.

item	value
specs	16 files in `../../e4/specs/` (`e4-cN.json`: 5 generators × 2000 `planted_shared` targets; `e4-cN-aa.json`: gstd, `target_seed` 20261020, 2000 targets)
protocol	`E0_E4.md` "E4 PROTOCOL", frozen in commit `b3eba557b` before the first record
command	`research/generator_effect_ic_20261019/e4/run_e4.sh`: `ecbench run --cpus none --allow-busy`, four sessions at a time, one lock file each
host	shared macOS host, 14 cores, about 46 load from other work; isolation L0 (no pinning, busy preflight allowed); wall time is not a metric
curves	c1–c8, 24-bit `a = −3` prime order (`../../e4/generators.jsonl`)
arms	`ic-abscissa` (`prime-abscissa:size=B`), `ic-knownlog-random` (`known-log:size=B,scalars=random`), both `mitm:negation_folded=1,m=2`, walk targets; B = ceil(r^(1/3))
seeds	measurement seed 20261019 (algorithm seed per workload and round); target seeds 20261019 (main) and 20261020 (A/A)
requested / completed	192 000 / 192 000 records (8 × 20 000 + 8 × 4 000), 192 000 verified, 0 failures, 0 timeouts
audit	`ecbench verify --replay 10` per session: `../../e4/verify.txt`
wall	22:15:49Z – 23:03:54Z (48 min); per session in `e4/times.txt` (main 875–1 164 s, A/A 162–365 s)
raw records	`e4/runs/` (packed as `../../e4/runs.tar.xz`, sha256 in `runs.tar.xz.sha256`)
aggregation	`examples/generator_effect_e4_analysis.rs` ⟶ `../../e4/analysis.json`

Measured values (all derived from exact operation counts)

The pooled ratio gstd / grand is the geometric mean over the 8 curves. The interval is a 90% bootstrap over runs within cells, 10 000 resamples.

arm	yield	S, w = 0	S, w = 0.1	S, w = 1
abscissa	0.9994 [0.9980, 1.0007]	1.0001 [0.9993, 1.0008]	1.0001 [0.9992, 1.0009]	1.0002 [0.9986, 1.0017]
known-log, random	0.9988 [0.9846, 1.0135]	1.0000 [1.0000, 1.0000]	1.0000 [1.0000, 1.0000]	1.0000 [0.9999, 1.0001]

The A/A ratio gstd / gstd-aa and the planted +3% control are in `analysis.json` and in E0_E4.md. Base construction and oracle set-up have one distinct value per curve and arm across all 12 000 records of that curve.

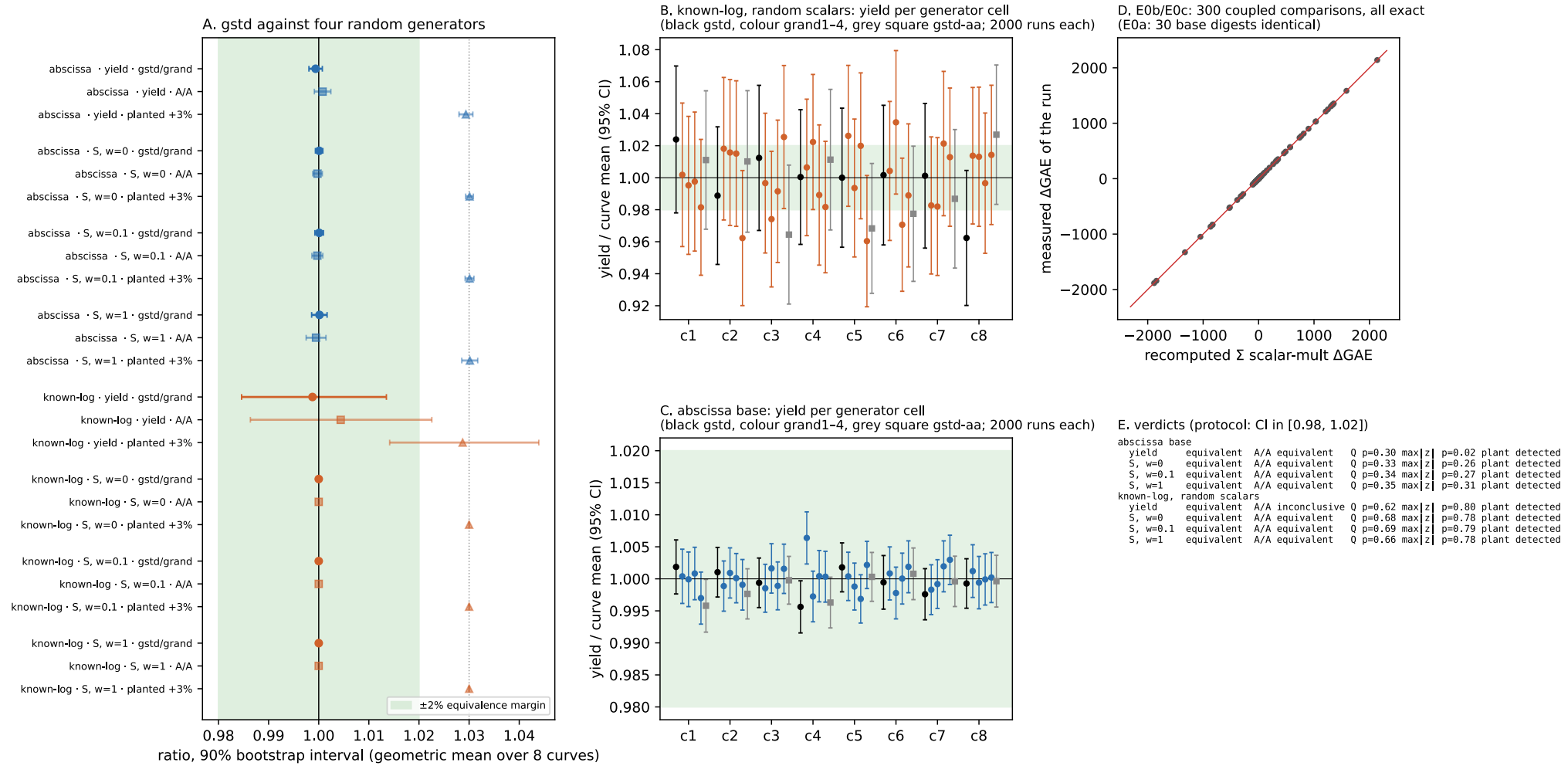


Figure 1: E4

Figure: the run's recorded values (`e4/plot_e4.py` from `e4/analysis.json` and `e0/e0_sweep.log`); sessions ECBS1h5482084cbe02, ECBS1h172c8ce5a162, ECBS1hc2d8241ad1ba, ECBS1h2db857bc6761, ECBS1h48a0812f6528, ECBS1he3de0a146b14, ECBS1h412a8daba919, ECBS1h3f5a2fdfff6d (main) and ECBS1hba211a0e071e, ECBS1h47e14b4b3342, ECBS1h70538d39d667, ECBS1hf2f98b50150c, ECBS1h70422a44cae0, ECBS1h6a8a38bf38a3, ECBS1h261d45f24f5c, ECBS1hbf40278ea7dc (A/A).